

Webinar 1 - R for Data Management

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Introduction

This R Markdown file is the **live-coding companion** to Webinar 1. It follows the session agenda below and uses a real **Malawi Demographic and Health Survey (MDHS)** Individual Recode (women’s) file throughout, so every function is demonstrated on genuine survey data rather than a toy example.

Session Setup

1. Install R

Download R from:

<https://cran.r-project.org/>

Steps: - Select your operating system (Windows/macOS/Linux). - Download and install the latest version.
- Accept the default installation settings.

2. Install RStudio

Download RStudio Desktop from:

<https://posit.co/download/rstudio-desktop/>

Steps: - Install RStudio after installing R. - Open RStudio and confirm that R is detected.

Research Focus

Determinants of Modern Contraceptive Use among Women Aged 15–49 Years

Modern contraceptive use is a key indicator of reproductive health and is essential for reducing unintended pregnancies, improving maternal and child health outcomes, and supporting sustainable population growth. Using data from the Malawi DHS, this session examines the determinants of modern contraceptive use among women aged 15–49, and how it varies by age, education, marital status, wealth, place of residence, region, employment, media exposure, and number of living children.

If you have downloaded your own DHS extract (`.dta` or `.sav`) from the DHS Program website, point `file_path` in the import step below to it.

Tip: DHS data requires free registration and project approval before download – start that process early if you plan to use your own country’s data.

R & RStudio Basics

How R Works

R has a set of methods and functions designed and packaged into libraries. For every programming task, you need to know the appropriate functions and packages to install and use; where a ready-made function doesn’t exist, you can write your own (a “user-defined function”).

RStudio is the IDE used to manage your datasets, scripts, and visualizations in one place.

Creating and Managing an R Project

Step 1: Create a New Project

- Open **RStudio**
- Go to **File -> New Project**
- Select **New Directory -> New Project**
- Enter a project name and choose a location
- Click **Create Project**

Step 2: Organize Your Project

Create folders to keep your work organized:

```
Project_Name/
|
|-- data/
|-- scripts/
|-- figures/
|-- output/
|-- reports/
'-- Project_Name.Rproj
```

Good Practices

- Use one project per analysis
- Organize files into folders
- Use relative file paths instead of hard-coded `setwd()` calls
- Save your work regularly
- Keep raw and cleaned data separate

Basic R Operations

R can handle everything from simple arithmetic to full statistical models. We will build up to the DHS analysis gradually, starting with the basics:

```
## the assignment operator
a <- 2
b <- 5:10
c <- c("a", "b", "d")
```

Loading Required Packages

```
# install.packages(c("tidyverse", "haven", "readxl", "labelled"))

library(tidyverse)  # dplyr, ggplot2, tidyr, stringr, etc.
library(haven)      # read Stata (.dta) / SPSS (.sav) files -- standard DHS formats
library(readxl)     # read Excel files
library(labelled)   # helper functions for labelled survey data
```

```
# Check and, if needed, set your working directory
getwd()
```

```
## [1] "C:/Users/FRANK/Desktop/R training/Data-analysis-in-R-2026"
```

```
# setwd("path/to/your/project")
```

Importing and Inspecting Data

Built in datasets

R comes with several datasets under the different libraries. Sometimes the research questions could be answered by some of these datasets.

```
# This functions shows you all available datasets under the base function  
  
# data()
```

Supported Data Formats

R can read data from many formats, including `.csv`, `.xlsx`, `.sav`, `.dta`, `.R/.RData`, and more. This session focuses on `.dta` (Stata format), since that is how the Malawi DHS Individual Recode file is distributed.

Importing the DHS Dataset

DHS data comes **already value-labelled**: every categorical variable (e.g., `v025` place of residence, `v106` education) carries its numeric code *and* a text label at the same time, via haven's `labelled` class. We keep both together rather than converting everything to plain factors on import, that way we still have the numeric codes available for `na_if()` and other numeric checks, but can pull in the ready-made labels with `as_factor()` whenever we want them (shown in **Working with Categorical Variables** below).

```
# Update this path to point to your own DHS extract  
# change these paths to make reference to the directory that has the dataset.  
file_path_dta <- "C:/Users/FRANK/Desktop/R training/DHS_data.dta"  
file_path_csv<- "C:/Users/FRANK/Desktop/R training/DHS_data.csv"  
  
dhs_data <- read_dta(file_path_dta)%>%  
  mutate(across(where(is.labelled), as_factor))  ### haven and convert values to value labels.
```

Note: we deliberately do *not* run `mutate(across(where(is.labelled), as_factor))` here. Doing so immediately would turn every categorical column into text labels straight away, which would break the numeric missing-value recoding (`na_if(education, 9)`, etc.) later in the Data Cleaning section. Converting variable-by-variable, only when we actually need the label, keeps both options available.

Selecting Variables of Interest

```
var_names <- c(  
  "caseid", # case id  
  "v005",   # sampling weight  
  "v021",   # primary sampling unit (cluster)  
  "v022",   # sampling strata  
  "v024",   # region  
  "v025",   # place of residence  
  "v012",   # age  
  "v106",   # highest education level
```

```

"v130", # religion
"v190", # wealth index
"v201", # children ever born
"v218", # number of living children
"v213", # currently pregnant
"v312", # current contraceptive method
"v313", # current contraceptive method type
"v501", # marital status
"v714", # currently working
"v717", # respondent's occupation
"v701", # husband's education
"v705", # husband's occupation
"v157", # reads newspaper
"v158", # listens to radio
"v159", # watches television
"v394", # visited health facility
"v481"  # health insurance coverage
)

dhs_raw <- dhs_data %>% select(var_names)

```

v005, v021, and v022 (the sampling weight, cluster, and strata) are not used in this session's basic descriptive analysis, but we keep them selected now – they're needed for **survey-weighted analysis**, which is previewed in Webinar 2.

First Look at the Data

```
dim(dhs_raw)
```

```
## [1] 24662    25
```

```
head(dhs_raw)
```

```

## # A tibble: 6 x 25
##   caseid      v005  v021 v022  v024  v025   v012 v106  v130  v190  v201  v218
##   <chr>      <dbl> <dbl> <fct> <fct> <fct> <dbl> <fct> <fct> <fct> <dbl> <dbl>
## 1 "      1 ~ 118748    1 ntch~ cent~ urban    30 seco~ ccap  rich~    3    2
## 2 "      1 ~ 118748    1 ntch~ cent~ urban    30 prim~ ccap  rich~    4    4
## 3 "      1 ~ 118748    1 ntch~ cent~ urban    28 prim~ othe~ rich~    3    3
## 4 "      1 ~ 118748    1 ntch~ cent~ urban    30 seco~ cath~ rich~    1    1
## 5 "      1 ~ 118748    1 ntch~ cent~ urban    16 seco~ othe~ rich~    0    0
## 6 "      1 ~ 118748    1 ntch~ cent~ urban    36 seco~ othe~ midd~    3    3
## # i 13 more variables: v213 <fct>, v312 <fct>, v313 <fct>, v501 <fct>,
## #   v714 <fct>, v717 <fct>, v701 <fct>, v705 <fct>, v157 <fct>, v158 <fct>,
## #   v159 <fct>, v394 <fct>, v481 <fct>

```

```
str(dhs_raw)
```

```
## tibble [24,662 x 25] (S3: tbl_df/tbl/data.frame)
```

```

## $ caseid: chr [1:24662] "      1 12 2" "      1 17 2" "      1 39 2" "      1 66 2" ...
##   .. attr(*, "label")= chr "case identification"
##   .. attr(*, "format.stata")= chr "%-9s"
## $ v005 : num [1:24662] 118748 118748 118748 118748 118748 ...
##   .. attr(*, "label")= chr "women's individual sample weight (6 decimals)"
##   .. attr(*, "format.stata")= chr "%10.0g"
## $ v021 : num [1:24662] 1 1 1 1 1 1 1 1 1 1 ...
##   .. attr(*, "label")= chr "primary sampling unit"
##   .. attr(*, "format.stata")= chr "%10.0g"
## $ v022 : Factor w/ 56 levels "chitipa - rural",...: 18 18 18 18 18 18 18 18 18 18 ...
##   .. attr(*, "label")= chr "sample strata for sampling errors"
## $ v024 : Factor w/ 3 levels "northern region",...: 2 2 2 2 2 2 2 2 2 2 ...
##   .. attr(*, "label")= chr "region"
## $ v025 : Factor w/ 2 levels "urban","rural": 1 1 1 1 1 1 1 1 1 1 ...
##   .. attr(*, "label")= chr "type of place of residence"
## $ v012 : num [1:24662] 30 30 28 30 16 36 24 23 28 29 ...
##   .. attr(*, "label")= chr "respondent's current age"
##   .. attr(*, "format.stata")= chr "%10.0g"
## $ v106 : Factor w/ 4 levels "no education",...: 3 2 2 3 3 3 3 3 3 3 ...
##   .. attr(*, "label")= chr "highest educational level"
## $ v130 : Factor w/ 8 levels "catholic","ccap",...: 2 2 5 1 5 5 2 2 5 2 ...
##   .. attr(*, "label")= chr "religion"
## $ v190 : Factor w/ 5 levels "poorest","poorer",...: 5 5 5 4 4 3 5 5 5 5 ...
##   .. attr(*, "label")= chr "wealth index combined"
## $ v201 : num [1:24662] 3 4 3 1 0 3 2 1 4 0 ...
##   .. attr(*, "label")= chr "total children ever born"
##   .. attr(*, "format.stata")= chr "%10.0g"
## $ v218 : num [1:24662] 2 4 3 1 0 3 2 1 4 0 ...
##   .. attr(*, "label")= chr "number of living children"
##   .. attr(*, "format.stata")= chr "%10.0g"
## $ v213 : Factor w/ 2 levels "no or unsure",...: 1 1 1 1 1 1 2 1 1 1 ...
##   .. attr(*, "label")= chr "currently pregnant"
## $ v312 : Factor w/ 21 levels "not using","pill",...: 1 12 12 4 1 1 1 12 4 1 ...
##   .. attr(*, "label")= chr "current contraceptive method"
## $ v313 : Factor w/ 4 levels "no method","folkloric method",...: 1 4 4 4 1 1 1 4 4 1 ...
##   .. attr(*, "label")= chr "current use by method type"
## $ v501 : Factor w/ 6 levels "never in union",...: 2 2 2 4 1 6 2 2 2 1 ...
##   .. attr(*, "label")= chr "current marital status"
## $ v714 : Factor w/ 2 levels "no","yes": 1 2 1 2 1 2 1 2 1 2 ...
##   .. attr(*, "label")= chr "respondent currently working"
## $ v717 : Factor w/ 11 levels "not working",...: 1 5 1 5 1 2 1 10 1 2 ...
##   .. attr(*, "label")= chr "respondent's occupation (grouped)"
## $ v701 : Factor w/ 5 levels "no education",...: 4 3 3 NA NA NA 3 3 3 NA ...
##   .. attr(*, "label")= chr "husband/partner's education level"
## $ v705 : Factor w/ 11 levels "did not work",...: 2 2 2 NA NA NA 10 2 2 NA ...
##   .. attr(*, "label")= chr "husband/partner's occupation (grouped)"
## $ v157 : Factor w/ 4 levels "not at all","less than once a week",...: 3 1 1 2 1 2 2 1 1 1 ...
##   .. attr(*, "label")= chr "frequency of reading newspaper or magazine"
## $ v158 : Factor w/ 4 levels "not at all","less than once a week",...: 3 3 1 2 1 2 3 3 3 2 ...
##   .. attr(*, "label")= chr "frequency of listening to radio"
## $ v159 : Factor w/ 4 levels "not at all","less than once a week",...: 3 3 1 2 1 1 2 3 1 2 ...
##   .. attr(*, "label")= chr "frequency of watching television"
## $ v394 : Factor w/ 2 levels "no","yes": 2 2 2 2 2 2 1 2 2 2 ...
##   .. attr(*, "label")= chr "visited health facility last 12 months"

```

```
## $ v481 : Factor w/ 2 levels "no","yes": 2 1 1 1 1 1 1 1 1 1 ...
## ..- attr(*, "label")= chr "covered by health insurance"
```

```
glimpse(dhs_raw)
```

```
## Rows: 24,662
## Columns: 25
## $ caseid <chr> "      1 12 2", "      1 17 2", "      1 39 2", " ~
## $ v005 <dbl> 118748, 118748, 118748, 118748, 118748, 118748, 118748, 118748, ~
## $ v021 <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ v022 <fct> ntchisi - urban, ntchisi - urban, ntchisi - urban, ntchisi - ur~
## $ v024 <fct> central region, central region, central region, central region, ~
## $ v025 <fct> urban, urban, urban, urban, urban, urban, urban, urban, urban, ~
## $ v012 <dbl> 30, 30, 28, 30, 16, 36, 24, 23, 28, 29, 22, 33, 26, 29, 15, 26, ~
## $ v106 <fct> secondary, primary, primary, secondary, secondary, secondary, s~
## $ v130 <fct> ccap, ccap, other christian, catholic, other christian, other c~
## $ v190 <fct> richest, richest, richest, richer, richer, middle, richest, ric~
## $ v201 <dbl> 3, 4, 3, 1, 0, 3, 2, 1, 4, 0, 2, 3, 2, 4, 0, 2, 1, 1, 2, 0, 0, ~
## $ v218 <dbl> 2, 4, 3, 1, 0, 3, 2, 1, 4, 0, 1, 3, 2, 4, 0, 2, 1, 1, 2, 0, 0, ~
## $ v213 <fct> no or unsure, no or unsure, no or unsure, no or unsure, no or u~
## $ v312 <fct> not using, implants/norplant, implants/norplant, injections, no~
## $ v313 <fct> no method, modern method, modern method, modern method, no meth~
## $ v501 <fct> married, married, married, widowed, never in union, no longer l~
## $ v714 <fct> no, yes, no, yes, no, yes, no, yes, no, yes, yes, yes, no, no, ~
## $ v717 <fct> not working, agricultural - self employed, not working, agricul~
## $ v701 <fct> higher, secondary, secondary, NA, NA, NA, secondary, secondary, ~
## $ v705 <fct> professional/technical/managerial, professional/technical/manag~
## $ v157 <fct> at least once a week, not at all, not at all, less than once a ~
## $ v158 <fct> at least once a week, at least once a week, not at all, less th~
## $ v159 <fct> at least once a week, at least once a week, not at all, less th~
## $ v394 <fct> yes, yes, yes, yes, yes, yes, no, yes, yes, yes, yes, yes, no, ~
## $ v481 <fct> yes, no, no, no, no, no, no, no, no, no, no, no, no, no, no, no~
```

```
summary(dhs_raw)
```

```
##      caseid          v005          v021          v022
## Length:24662      Min.   : 12778      Min.   : 1.0      mangochi - rural: 863
## Class :character  1st Qu.: 428519      1st Qu.:215.0      thyolo - rural  : 830
## Mode  :character  Median : 854425      Median :425.0      dedza - rural   : 826
##                  Mean   :1000052      Mean   :425.4      dowa - rural    : 820
##                  3rd Qu.:1289055      3rd Qu.:635.0      ntcheu - rural  : 807
##                  Max.   :6400914      Max.   :850.0      phalombe - rural: 807
##                                     (Other)      :19709
##
##      v024          v025          v012          v106
## northern region: 4829      urban: 5267      Min.   :15.00      no education: 2786
## central region : 8454      rural:19395      1st Qu.:20.00      primary      :15090
## southern region:11379                                     Median :27.00      secondary   : 6085
##                                                         Mean   :28.13      higher      : 701
##                                                         3rd Qu.:35.00
##                                                         Max.   :49.00
##
##
##      v130          v190          v201
## other christian          :10436      poorest:4295      Min.   : 0.000
```



```

## catholic                : 4337    poorer :4445    1st Qu.: 1.000
## ccap                    : 3894    middle :4530    Median : 2.000
## muslim                  : 2739    richer :4916    Mean   : 2.769
## seventh day adventist / baptist: 1843    richest:6476    3rd Qu.: 4.000
## anglican                : 1262                                Max.   :15.000
## (Other)                 : 151
##      v218                v213                v312
## Min.   : 0.000    no or unsure:22822    not using      :13426
## 1st Qu.: 1.000    yes           : 1840    injections     : 5445
## Median : 2.000                                implants/norplant : 2328
## Mean   : 2.475                                female sterilization: 1900
## 3rd Qu.: 4.000                                male condom     : 679
## Max.   :11.000                                pill            : 421
##                                           (Other)        : 463
##      v313                v501
## no method      :13426    never in union      : 5351
## folkloric method : 53    married            :14977
## traditional method: 153    living with partner : 1041
## modern method   :11030    widowed            : 743
##                                           divorced           : 1240
##                                           no longer living together/separated: 1310
##
##      v714                v717                v701
## no : 9542    agricultural - self employed :9181    no education:1460
## yes:15120    not working :8456    primary :8408
##                                           unskilled manual :3122    secondary :5033
##                                           professional/technical/managerial:1341    higher : 921
##                                           sales :1136    don't know : 196
##                                           skilled manual : 363    NA's :8644
##                                           (Other) :1063
##
##      v705                v157
## agricultural - self employed :5413    not at all :19481
## unskilled manual :2988    less than once a week: 3024
## skilled manual :2212    at least once a week : 2157
## professional/technical/managerial:1563    almost every day : 0
## did not work :1494
## (Other) :2348
## NA's :8644
##
##      v158                v159                v394
## not at all :12365    not at all :19144    no : 9440
## less than once a week: 4248    less than once a week: 2117    yes:15222
## at least once a week : 8049    at least once a week : 3401
## almost every day : 0    almost every day : 0
##
##
##
##      v481
## no :24325
## yes: 337
##
##
##
##
##

```

Inspecting the Embedded Value Labels

Since this data comes already value-labelled, it's worth checking what labels each variable actually carries *before* deciding how to recode it – this tells us whether we can use `as_factor()` directly, or whether we still need to define labels ourselves.

```
# Quick look at two key variables
val_labels(dhs_raw$v025)
```

```
## NULL
```

```
val_labels(dhs_raw$v106)
```

```
## NULL
```

```
# A full data-dictionary-style overview of every variable at once
# look_for(dhs_raw)
```

Data Cleaning – Part 1

Objectives

By the end of this segment, participants should be able to:

- Rename variables
- Identify and handle missing values
- Detect and remove duplicate records
- Recode variables
- Work with categorical variables
- Detect and handle outliers

Renaming Variables

DHS datasets use standardized variable names (e.g., `v012`, `v106`). Renaming them up front makes every subsequent step easier to read.

```
dhs <- dhs_raw %>%
  rename(
    age           = v012,
    region        = v024,
    residence     = v025,
    education     = v106,
    religion      = v130,
    wealth_quintile = v190,
    children_ever_born = v201,
    pregnant      = v213,
    living_children = v218,
```

```

contraceptive      = v313,
marital_status     = v501,
currently_working  = v714,
occupation         = v717,
husband_education  = v701,
husband_occupation = v705,
reads_newspaper    = v157,
listens_radio      = v158,
watches_tv         = v159,
visited_health_facility = v394,
health_insurance   = v481
)

```

```
names(dhs)
```

```

## [1] "caseid"          "v005"
## [3] "v021"            "v022"
## [5] "region"          "residence"
## [7] "age"             "education"
## [9] "religion"        "wealth_quintile"
## [11] "children_ever_born" "living_children"
## [13] "pregnant"        "v312"
## [15] "contraceptive"    "marital_status"
## [17] "currently_working" "occupation"
## [19] "husband_education" "husband_occupation"
## [21] "reads_newspaper"  "listens_radio"
## [23] "watches_tv"       "visited_health_facility"
## [25] "health_insurance"

```

Identifying Missing Values

```

# Count missing values for every column
colSums(is.na(dhs))

```

```

##           caseid           v005           v021
##           0           0           0
##           v022           region           residence
##           0           0           0
##           age           education           religion
##           0           0           0
##           wealth_quintile children_ever_born living_children
##           0           0           0
##           pregnant           v312           contraceptive
##           0           0           0
##           marital_status currently_working           occupation
##           0           0           0
##           husband_education husband_occupation reads_newspaper
##           8644           8644           0
##           listens_radio           watches_tv visited_health_facility
##           0           0           0
##           health_insurance
##           0

```

Handling Missing Values

DHS commonly uses special numeric codes for “missing”/“don’t know” instead of a blank cell. We convert these to proper NA so R treats them correctly.

```
colSums(is.na(dhs))
```

```
##           caseid           v005           v021
##           0           0           0
##           v022           region           residence
##           0           0           0
##           age           education           religion
##           0           0           0
##           wealth_quintile children_ever_born living_children
##           0           0           0
##           pregnant           v312           contraceptive
##           0           0           0
##           marital_status currently_working           occupation
##           0           0           0
##           husband_education husband_occupation reads_newspaper
##           8644           8644           0
##           listens_radio           watches_tv visited_health_facility
##           0           0           0
##           health_insurance
##           0
```

```
# Let's check the husband_education & husband_occupation

# dhs %>% filter(is.na(husband_education)) %>% dim
# dhs %>% filter(is.na(husband_education)) %>% view() ### sometimes missingness is a design intent not

# Muted since there are no missing ness.
# dhs <- dhs %>%
#   mutate(
#     age           = na_if(age, 98),
#     education      = na_if(education, 9),
#     pregnant       = na_if(pregnant, 9),
#     living_children = na_if(living_children, 99)
#   )
```

Removing Duplicate Records

Duplicate records can give a misleading picture of the data by over-weighting a small number of respondents.

```
# How many caseids appear more than once?
sum(duplicated(dhs$caseid))
```

```
## [1] 100
```

```
# Inspect the duplicated rows themselves (both occurrences)
dhs_duplicates <- dhs %>%
```

```

filter(duplicated(caseid) | duplicated(caseid, fromLast = TRUE))

dhs_duplicates

## # A tibble: 200 x 25
##   caseid      v005 v021 v022 region residence age education religion
##   <chr>      <dbl> <dbl> <fct> <fct> <fct>   <dbl> <fct>   <fct>
## 1 "        6 97 ~ 2.01e6    6 mzim~ north~ rural    35 primary other c~
## 2 "       23 42 ~ 6.93e5   23 mang~ south~ rural    21 secondary other c~
## 3 "       27 159 ~ 4.91e5   27 rump~ north~ rural    31 primary catholic
## 4 "       27 196 ~ 4.91e5   27 rump~ north~ rural    34 primary other c~
## 5 "       33 42 ~ 1.04e6   33 dowa~ centr~ rural    19 primary ccap
## 6 "       37 3 ~ 8.20e5   37 mzim~ north~ urban    29 secondary catholic
## 7 "       38 10 ~ 2.31e5   38 mchi~ centr~ urban    22 primary other c~
## 8 "       40 100 ~ 3.34e5   40 nkha~ north~ rural    40 secondary ccap
## 9 "       64 209 ~ 1.50e6   64 zomb~ south~ rural    15 primary muslim
## 10 "      83 49 ~ 2.70e5   83 neno~ south~ rural    22 secondary catholic
## # i 190 more rows
## # i 16 more variables: wealth_quintile <fct>, children_ever_born <dbl>,
## #   living_children <dbl>, pregnant <fct>, v312 <fct>, contraceptive <fct>,
## #   marital_status <fct>, currently_working <fct>, occupation <fct>,
## #   husband_education <fct>, husband_occupation <fct>, reads_newspaper <fct>,
## #   listens_radio <fct>, watches_tv <fct>, visited_health_facility <fct>,
## #   health_insurance <fct>

```

```

# Keep only the first occurrence of each caseid
dhs <- dhs %>% distinct(caseid, .keep_all = TRUE)
nrow(dhs)

```

```
## [1] 24562
```

Recoding Variables

`residence` and `education` are already value-labelled by DHS, so we don't need to manually retype their coding scheme – that comes in the next section via `as_factor()`. The one variable that genuinely needs manual recoding here is `age_group`, since it's a brand-new category that doesn't exist as a raw DHS variable and so has no label to inherit:

```

dhs <- dhs %>%
  mutate(
    age_group = case_when(
      age >= 15 & age < 20 ~ "15-19",
      age >= 20 & age < 35 ~ "20-34",
      age >= 35 & age <= 49 ~ "35-49",
      TRUE ~ NA_character_
    )
  )
# An equivalent (but harder to read for 3+ categories) version with
# nested ifelse() would be:
# age_group = ifelse(age >= 15 & age < 20, "15-19",
#   ifelse(age >= 20 & age < 35, "20-34",
#     ifelse(age >= 35 & age <= 49, "35-49", NA)))

```

```
)

count(dhs, age_group)
```

```
## # A tibble: 3 x 2
##   age_group      n
##   <chr>      <int>
## 1 15-19      5273
## 2 20-34     12718
## 3 35-49      6571
```

Working with Categorical Variables (Factors)

Because `residence` and `education` are already value-labelled, `as_factor()` pulls in DHS's own text labels directly – no manual mapping required, and no risk of a typo creating a mismatched category:

```
dhs <- dhs %>%
  mutate(
    residence_f = as_factor(residence),
    education_f = factor(as_factor(education), ordered = TRUE),
    age_group_f = factor(age_group, levels = c("15-19", "20-34", "35-49"))
  )

levels(dhs$residence_f)
```

```
## [1] "urban" "rural"
```

```
levels(dhs$education_f)
```

```
## [1] "no education" "primary"      "secondary"    "higher"
```

```
table(dhs$education_f, useNA = "ifany")
```

```
##
## no education      primary      secondary      higher
##           2779         15028         6061         694
```

`as_factor()` gives us the categories DHS already defined. Manual recoding still earns its keep when you want *different* groupings than the source labels provide – for example, collapsing four education levels down to two for a simpler comparison:

```
dhs <- dhs %>%
  mutate(
    education_grouped = fct_collapse(
      education_f,
      "No education" = "No education",
      "Primary or higher" = c("Primary", "Secondary", "Higher")
    )
  )

table(dhs$education_grouped)
```

```
##
## no education      primary      secondary      higher
##           2779           15028           6061           694
```

Outlier Detection and Removal

Outliers are values that fall far outside a variable's typical range. Left unaddressed, they can distort means, standard deviations, and any models built on the data. Two complementary checks are useful:

1. **Validity / range checks** – for variables with a known plausible range. This survey only interviews women aged 15–49, so any value outside that range is a data error, not a genuine outlier.
2. **Statistical outlier detection** – for continuous variables without a fixed valid range (e.g., number of living children), where a rule such as the interquartile range (IQR) flags unusually extreme values.

Validity check: age

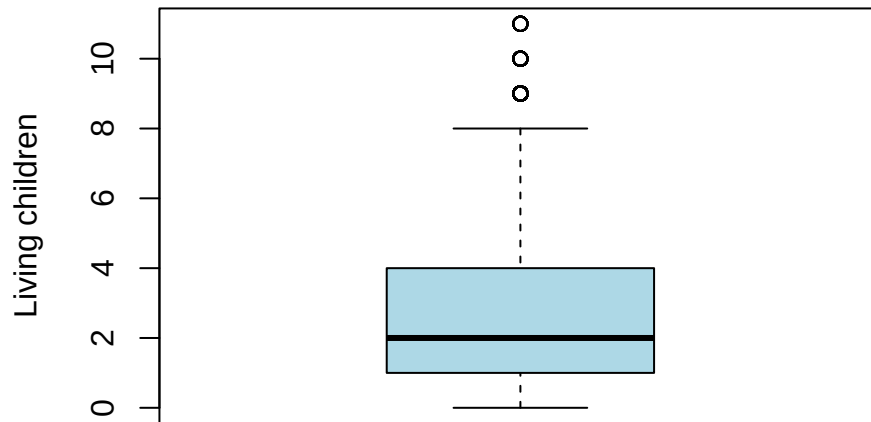
```
# This should return 0 rows if age was recorded correctly
dhs %>% filter(age < 15 | age > 49)
```

```
## # A tibble: 0 x 30
## # i 30 variables: caseid <chr>, v005 <dbl>, v021 <dbl>, v022 <fct>,
## #   region <fct>, residence <fct>, age <dbl>, education <fct>, religion <fct>,
## #   wealth_quintile <fct>, children_ever_born <dbl>, living_children <dbl>,
## #   pregnant <fct>, v312 <fct>, contraceptive <fct>, marital_status <fct>,
## #   currently_working <fct>, occupation <fct>, husband_education <fct>,
## #   husband_occupation <fct>, reads_newspaper <fct>, listens_radio <fct>,
## #   watches_tv <fct>, visited_health_facility <fct>, ...
```

Statistical outlier detection: living children

```
boxplot(dhs$living_children,
        main = "Number of Living Children",
        ylab = "Living children",
        col = "lightblue")
```

Number of Living Children



```
Q1      <- quantile(dhs$living_children, 0.25, na.rm = TRUE)
Q3      <- quantile(dhs$living_children, 0.75, na.rm = TRUE)
IQR_val <- IQR(dhs$living_children, na.rm = TRUE)
```

```
lower_bound <- Q1 - 1.5 * IQR_val
upper_bound <- Q3 + 1.5 * IQR_val
```

```
c(lower_bound = lower_bound, upper_bound = upper_bound)
```

```
## lower_bound.25% upper_bound.75%
##           -3.5           8.5
```

```
# How many observations fall outside these bounds?
```

```
dhs %>%
  filter(living_children < lower_bound | living_children > upper_bound) %>%
  nrow()
```

```
## [1] 159
```

```
# Flag outliers rather than silently deleting them, so the decision stays
# auditable
```

```
dhs <- dhs %>%
  mutate(
    living_children_outlier = living_children < lower_bound |
                             living_children > upper_bound
  )
```

```
table(dhs$living_children_outlier, useNA = "ifany")
```



```
##
## FALSE TRUE
## 24403 159

# Remove confirmed outliers (missing values were already handled earlier,
# so we keep NAs rather than dropping them here)
dhs <- dhs %>%
  filter(is.na(living_children) | !living_children_outlier)
```

Note: removing outliers is not always the right choice – a woman may genuinely have an unusually high number of children. Always inspect flagged cases before deleting them, and document the rule you used.

Data Cleaning – Part 2

Objectives

Participants should be able to:

- Create new variables using `mutate()`
- Select variables using `select()`
- Filter observations using `filter()`
- Arrange data using `arrange()`
- Combine cleaning steps into a single pipeline

Creating New Variables with `mutate()`

The DHS variable `contraceptive` (originally `v313`) classifies contraceptive methods as:

Code	Method Type
0	No method
1	Folkloric
2	Traditional
3	Modern

```
dhs <- dhs %>%
  mutate(modern_use = ifelse(contraceptive == "modern method", "yes", "no"))
head(dhs$modern_use)
```

```
## [1] "no" "yes" "yes" "yes" "no" "no"
```

We also create a media exposure indicator for women exposed to at least one media source:

```
dhs <- dhs %>%
  mutate(
    media_exposure = ifelse(
      reads_newspaper != "not at all" | listens_radio != "not at all" | watches_tv != "not at all",
      "Yes", "No"
    )
  )

table(dhs$media_exposure, useNA = "ifany")
```

```
##
##      No    Yes
## 10492 13911
```

Selecting Variables with `select()`

General syntax:

```
data %>% select(variable1, variable2)
```

Select a few variables to preview:

```
dhs %>%
  select(age, education_f, residence_f, modern_use) %>%
  head()
```

```
## # A tibble: 6 x 4
##   age education_f residence_f modern_use
##   <dbl> <ord>      <fct>      <chr>
## 1    30 secondary  urban      no
## 2    30 primary   urban      yes
## 3    28 primary   urban      yes
## 4    30 secondary urban      yes
## 5    16 secondary urban      no
## 6    36 secondary urban      no
```

Remove variables you no longer need (e.g., the individual media items, once `media_exposure` has been derived from them):

```
dhs %>%
  select(-reads_newspaper, -listens_radio, -watches_tv) %>%
  names()
```

```
## [1] "caseid"          "v005"
## [3] "v021"            "v022"
## [5] "region"          "residence"
## [7] "age"             "education"
## [9] "religion"        "wealth_quintile"
## [11] "children_ever_born" "living_children"
## [13] "pregnant"        "v312"
```

```
## [15] "contraceptive"      "marital_status"
## [17] "currently_working"  "occupation"
## [19] "husband_education"  "husband_occupation"
## [21] "visited_health_facility" "health_insurance"
## [23] "age_group"          "residence_f"
## [25] "education_f"         "age_group_f"
## [27] "education_grouped"   "living_children_outlier"
## [29] "modern_use"          "media_exposure"
```

Select by name pattern – most useful *before* renaming, directly on the raw DHS codes:

```
# Variables starting with "v1"
dhs_raw %>% select(starts_with("v1")) %>% names()
```

```
## [1] "v106" "v130" "v190" "v157" "v158" "v159"
```

```
# Variables ending in "01"
dhs_raw %>% select(ends_with("01")) %>% names()
```

```
## [1] "v201" "v501" "v701"
```

Filtering Observations with filter()

General syntax:

```
data %>% filter(condition)
```

```
# Women aged 15-49
women_15_49 <- dhs %>% filter(age >= 15 & age <= 49)

# Rural women
rural <- dhs %>% filter(residence == "rural")

# Urban women
urban <- dhs %>% filter(residence == "urban")

# Married women
married <- dhs %>% filter(marital_status == "married")

# Women using a modern contraceptive method
users <- dhs %>% filter(modern_use == "yes")

# Multiple conditions at once: rural, secondary+ education, modern users
subset_data <- dhs %>%
  filter(residence == 'urban', education %in% c("secondary", "higher"), modern_use == "yes")

nrow(women_15_49); nrow(rural); nrow(urban); nrow(married); nrow(users); nrow(subset_data)

## [1] 24403

## [1] 19160
```

```
## [1] 5243

## [1] 14783

## [1] 10920

## [1] 0
```

Arranging Data with arrange()

```
# Youngest to oldest
dhs %>% arrange(age) %>% select(caseid, age) %>% head()
```

```
## # A tibble: 6 x 2
##   caseid      age
##   <chr>      <dbl>
## 1 "      1 178 3"    15
## 2 "      1 290 4"    15
## 3 "      2  15 5"    15
## 4 "      2 205 5"    15
## 5 "      4  16 9"    15
## 6 "      4 171 2"    15
```

```
# Oldest to youngest
dhs %>% arrange(desc(age)) %>% select(caseid, age) %>% head()
```

```
## # A tibble: 6 x 2
##   caseid      age
##   <chr>      <dbl>
## 1 "      3 167 1"    49
## 2 "      3 192 2"    49
## 3 "      5 104 1"    49
## 4 "      7  26 2"    49
## 5 "      7 117 1"    49
## 6 "     10  62 2"    49
```

Putting It All Together

The steps above were shown one at a time for teaching purposes. In practice, you would chain them into a single reproducible pipeline. This recap starts again from the **raw, unrenamed** variable selection (`dhs_raw`) and produces the final analysis dataset we will use for the rest of this session – reusing the same outlier bounds computed earlier:

```
analysis_data <- dhs_raw %>%
  rename(
    age      = v012,
    region   = v024,
    residence = v025,
    education = v106,
```

```

religion      = v130,
wealth_quintile = v190,
children_ever_born = v201,
pregnant      = v213,
living_children = v218,
contraceptive  = v313,
marital_status = v501,
reads_newspaper = v157,
listens_radio  = v158,
watches_tv     = v159
) %>%
distinct(caseid, .keep_all = TRUE) %>%
mutate(
  residence_f = as_factor(residence),
  education_f = factor(as_factor(education), ordered = TRUE),
  age_group_f = factor(case_when(
    age >= 15 & age < 20 ~ "15-19",
    age >= 20 & age < 35 ~ "20-34",
    age >= 35 & age <= 49 ~ "35-49",
    TRUE ~ NA_character_),
    levels = c("15-19", "20-34", "35-49")),
  modern_use = ifelse(contraceptive == "modern method", "yes", "no"),
  media_exposure = ifelse(reads_newspaper != "not at all" | listens_radio != "not at all" | watches_tv != "not at all", "yes", "no"),
  filter(is.na(living_children) |
    !(living_children < lower_bound | living_children > upper_bound)) %>%
  filter(age >= 15 & age <= 49) %>%
  select(caseid, age, age_group_f, residence_f, education_f, wealth_quintile,
    region, marital_status, children_ever_born, living_children,
    modern_use, media_exposure)

glimpse(analysis_data)

```

```

## Rows: 24,403
## Columns: 12
## $ caseid      <chr> "      1 12 2", "      1 17 2", "      1 39~
## $ age         <dbl> 30, 30, 28, 30, 16, 36, 24, 23, 28, 29, 22, 33, 26,~
## $ age_group_f <fct> 20-34, 20-34, 20-34, 20-34, 15-19, 35-49, 20-34, 20~
## $ residence_f <fct> urban, urban, urban, urban, urban, urban, urban, ur~
## $ education_f <ord> secondary, primary, primary, secondary, secondary, ~
## $ wealth_quintile <fct> richest, richest, richest, richer, richer, middle, ~
## $ region       <fct> central region, central region, central region, cen~
## $ marital_status <fct> married, married, married, widowed, never in union,~
## $ children_ever_born <dbl> 3, 4, 3, 1, 0, 3, 2, 1, 4, 0, 2, 3, 2, 4, 0, 2, 1, ~
## $ living_children <dbl> 2, 4, 3, 1, 0, 3, 2, 1, 4, 0, 1, 3, 2, 4, 0, 2, 1, ~
## $ modern_use   <chr> "no", "yes", "yes", "yes", "no", "no", "no", "yes",~
## $ media_exposure <chr> "Yes", "Yes", "No", "Yes", "No", "Yes", "Yes", "Yes~

```

analysis_data is now our single, clean, analysis-ready dataset. Every descriptive statistic from here on uses it.

Descriptive Statistics

Objectives

Participants should be able to:

- Generate frequencies
- Calculate means and medians
- Produce percentages
- Create summary tables

Numerical Summaries

```
mean(analysis_data$age, na.rm = TRUE)
```

```
## [1] 28.03508
```

```
median(analysis_data$age, na.rm = TRUE)
```

```
## [1] 27
```

```
range(analysis_data$age, na.rm = TRUE)
```

```
## [1] 15 49
```

```
sd(analysis_data$age, na.rm = TRUE)
```

```
## [1] 9.193104
```

```
quantile(analysis_data$age, na.rm = TRUE)
```

```
##   0%   25%   50%   75%  100%
```

```
##   15    20    27    35    49
```

```
summary(analysis_data$age)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```

```
##   15.00   20.00   27.00   28.04   35.00   49.00
```

Frequency Tables

```
table(analysis_data$education_f)
```

```
##
```

```
## no education      primary      secondary      higher
```

```
##          2706          14946          6057          694
```

```
table(analysis_data$residence_f)
```

```
##  
## urban rural  
## 5243 19160
```

```
table(analysis_data$modern_use)
```

```
##  
## no yes  
## 13483 10920
```

Relative Frequencies and Percentages

```
prop.table(table(analysis_data$education_f))
```

```
##  
## no education primary secondary higher  
## 0.11088801 0.61246568 0.24820719 0.02843913
```

```
round(prop.table(table(analysis_data$education_f)) * 100, 1)
```

```
##  
## no education primary secondary higher  
## 11.1 61.2 24.8 2.8
```

Two-Way Tables

```
# Education by contraceptive use  
table(analysis_data$education_f, analysis_data$modern_use)
```

```
##  
## no yes  
## no education 1441 1265  
## primary 7955 6991  
## secondary 3651 2406  
## higher 436 258
```

```
# Row percentages (within each education level)  
round(prop.table(table(analysis_data$education_f,  
analysis_data$modern_use), 1) * 100, 1)
```

```
##  
## no yes  
## no education 53.3 46.7  
## primary 53.2 46.8  
## secondary 60.3 39.7  
## higher 62.8 37.2
```

```
# Column percentages (within each contraceptive-use group)
round(prop.table(table(analysis_data$education_f,
                      analysis_data$modern_use), 2) * 100, 1)
```

```
##
##           no  yes
## no education 10.7 11.6
## primary      59.0 64.0
## secondary    27.1 22.0
## higher        3.2  2.4
```

Descriptive Statistics Using dplyr

```
analysis_data %>%
  summarise(
    Mean_Age = mean(age, na.rm = TRUE),
    Median_Age = median(age, na.rm = TRUE),
    SD_Age = sd(age, na.rm = TRUE),
    Minimum = min(age, na.rm = TRUE),
    Maximum = max(age, na.rm = TRUE)
  )
```

```
## # A tibble: 1 x 5
##   Mean_Age Median_Age SD_Age Minimum Maximum
##   <dbl>      <dbl> <dbl>    <dbl>    <dbl>
## 1    28.0         27  9.19      15      49
```

Grouped Summaries

```
analysis_data %>%
  group_by(residence_f) %>%
  summarise(
    Women = n(),
    Mean_Age = mean(age, na.rm = TRUE),
    Median_Age = median(age, na.rm = TRUE)
  )
```

```
## # A tibble: 2 x 4
##   residence_f Women Mean_Age Median_Age
##   <fct>      <int>    <dbl>    <dbl>
## 1 urban      5243    27.6      26
## 2 rural     19160    28.2      27
```

```
analysis_data %>%
  group_by(education_f) %>%
  summarise(
    Women = n(),
    Mean_Age = mean(age, na.rm = TRUE),
    Modern_Use = mean(modern_use, na.rm = TRUE) * 100
  )
```



```
## # A tibble: 4 x 4
##   education_f Women Mean_Age Modern_Use
##   <ord>      <int>    <dbl>    <dbl>
## 1 no education  2706    35.4      NA
## 2 primary     14946    27.7      NA
## 3 secondary    6057    25.4      NA
## 4 higher       694     29.7      NA
```

```
analysis_data %>%
  group_by(residence_f, education_f) %>%
  summarise(
    Women      = n(),
    Mean_Age   = mean(age, na.rm = TRUE),
    Modern_Use = mean(modern_use, na.rm = TRUE) * 100,
    .groups = "drop"
  )
```

```
## # A tibble: 8 x 5
##   residence_f education_f Women Mean_Age Modern_Use
##   <fct>      <ord>      <int>    <dbl>    <dbl>
## 1 urban      no education   217    35.3      NA
## 2 urban      primary     2079    28.4      NA
## 3 urban      secondary    2461    25.8      NA
## 4 urban      higher       486    29.9      NA
## 5 rural      no education  2489    35.4      NA
## 6 rural      primary     12867    27.6      NA
## 7 rural      secondary    3596    25.2      NA
## 8 rural      higher       208    29.3      NA
```

Simple Summary Tables for Reporting

```
analysis_data %>% count(residence_f)
```

```
## # A tibble: 2 x 2
##   residence_f n
##   <fct>      <int>
## 1 urban      5243
## 2 rural     19160
```

```
analysis_data %>% count(education_f)
```

```
## # A tibble: 4 x 2
##   education_f n
##   <ord>      <int>
## 1 no education 2706
## 2 primary     14946
## 3 secondary    6057
## 4 higher       694
```

```
analysis_data %>% count(residence_f, modern_use)
```

```
## # A tibble: 4 x 3
##   residence_f modern_use     n
##   <fct>      <chr>      <int>
## 1 urban      no           3008
## 2 urban      yes           2235
## 3 rural      no          10475
## 4 rural      yes           8685
```

Practical Exercise

Using the Malawi DHS dataset:

1. Import the dataset and select the variables of interest.
 2. Rename variables to meaningful names.
 3. Identify and handle missing values.
 4. Check for and remove duplicate records.
 5. Recode residence, education, and age into meaningful categories, and convert them to factors.
 6. Detect and handle outliers in the number of living children using the IQR method.
 7. Create a binary variable for modern contraceptive use and a media exposure indicator.
 8. Filter the data to women aged 15–49 and select the final variables needed for analysis.
 9. Compute the mean and median age.
 10. Produce frequency tables for education and residence.
 11. Generate a cross-tabulation of education and modern contraceptive use, with row percentages.
 12. Summarize the number of women and mean age by education level using `dplyr`.
-

Key Takeaways

By the end of this practical, participants should be able to:

- Import and inspect DHS datasets in R
- Clean and prepare data for analysis
- Rename and recode variables
- Handle missing values
- Detect and handle outliers using visual and statistical methods
- Create new variables using `mutate()`
- Filter and select data using `dplyr`
- Produce descriptive statistics and summary tables